

Thrust & Drift — Field Manual

Version 1.20.0 · Mongoose Traveller 2e Space Combat Simulator

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1. Overview

Thrust & Drift is a browser-based Virtual Tabletop (VTT) tool for running Mongoose Traveller 2nd Edition space combat at the gaming table. It is GM-operated and designed for shared-screen play — one person drives, everyone watches.

Two combat modes are available:

Mode	Description
VECTORIAL	Full hex grid with velocity vectors. Ships move according to their accumulated velocity each round. Implements <i>Traveller Companion 2024</i> pp. 169–186.
BASIC	Simplified range-band system. No hex map, no vectors. Faster for small engagements.

Note: The mode is selected on the Dashboard before starting a session and cannot be changed mid-battle.

2. Dashboard

The Dashboard is the pre-battle lobby. It has two columns: the **Operations Console** on the left and **Ship Profiles** on the right.

2.1 Ship Profiles

The right panel lists all saved ship profiles.

Control	Action
+ NEW PROFILE	Create a new ship from scratch.
 Edit	Modify an existing profile.
 Duplicate	Clone a profile as a starting point.
 Delete	Remove a profile (confirmation required).
 IMPORT	Load profiles from a <code>.json</code> file.
 EXPORT	Save all current profiles to a <code>.json</code> file.

Control	Action
 CATALOG	Browse the built-in High Guard 2022 catalog and add ships to your profiles.

2.2 Operations Console

Control	Action
COMBAT MODE	Toggle between Vectorial and Basic before starting.
 RESUME	Appears when an autosaved session is found. Click to restore instantly. The right panel shows the full saved roster (round, phase, ships by faction with hull bars).
×	Clears the autosave from IndexedDB. Appears next to  RESUME .
▶ NEW SESSION	Clears any existing battle state and enters the combat map.
↓ RESUME FROM FILE	Load a previously saved <code>.json</code> session. A preview screen shows the full roster before you confirm.
 FIELD MANUAL	Opens this manual inside the app.
 CHANGELOG	Opens the in-app version history (all releases with change categories).

2.3 Ship Profile Form

Each profile stores the ship's stats and named crew members.

Field	Description
Name / Tonnage	Display name and hull size (affects target size DM).
Hull / Thrust	Max hull points and base manoeuvre drive rating.
Tech Level (TL)	Ship's technology level (default 12). Gates Smart guidance on all missile and torpedo weapons — the DM+2 Smart bonus applies only when the <i>launching</i> ship's TL ≥ 9 (CRB p.79). Set to 8 or below for pre-stellar opponents or salvaged/primitive vessels. Most standard Traveller-era ships are TL 12–15.
Turrets	Add weapon turrets: type, damage dice, range band, and special traits. Maximum 3 weapons per turret (triple turret, CRB p.163).
Crew	Named crew members, each with individual skill levels (see § 11).

***Design note — why TL on the ship, not the weapon:** CRB p.79 states that Smart munitions require a fire control system at TL9+ to guide them. All missile and torpedo weapons already carry the **Smart** trait in the weapon table — this correctly represents the munition's capability. The TL field on the ship represents whether the vessel's targeting computer can actually exploit that capability. Separating the two avoids silent incorrect DMs when early-TL or pirate vessels appear alongside standard Imperial fleet ships.*

3. Map Controls

The battle map is a **flat-top hex grid**. All interaction is mouse-driven.

Input	Action
Left-drag	Pan the map.
Scroll wheel	Zoom in / out.

Input	Action
Double-click	Centre the map on that hex.
Left-click token	Select the ship (highlights it).
Right-click hex	Open context menu — actions depend on the hex content and the current phase.
 (HUD)	Audio toggle — enables or mutes procedural sound effects (laser fire, impacts, thrust plume, missile launch). No audio files required.

Note: Context menu actions are phase-gated **and initiative-gated**. Only options valid for the current phase are shown — and in the Acceleration, Attack, and Actions phases, combat actions are shown only for the ship whose turn it currently is. Right-clicking another ship shows "Not this ship's turn".

3.1 Legend

Click  **Legend** (fixed button, top-right of the battle screen) to open the visual reference panel. Also accessible via right-click any **empty hex** → **Legend**.

Category	Symbols
Tokens	Ship silhouette (6 shapes: delta, needle, freighter, gunship, cruiser, capital — each rotates to face velocity direction); HP arc (green/yellow/red); missile salvo (three staggered yellow silhouettes — rotates to face velocity direction; count + thrust arc shown; hover for launcher/target/thrust tooltip); torpedo (red/amber silhouette — separate salvo type)
Turret beams	Pulse Laser (sky blue), Beam Laser (blue), Particle Beam (purple), Railgun (orange), Fusion Gun (amber-white), Plasma Gun (magenta), Ion Cannon (electric blue — no hull damage, applies thrust penalty)

Category	Symbols
Barbette beams	Pulse Laser Barbette (sky blue, thicker), Beam Laser Barbette (blue, thicker), Particle Barbette (purple, thicker), Fusion Barbette (amber-white, thicker), Plasma Barbette (magenta, thicker), Railgun Barbette (orange, thicker) — all barbettes deal ×3 damage after armour
Hit effects	Impact burst (expanding sparks on target), Critical flash (red ring + label), Ion burst (blue ring — Ion Cannon hit)
Movement effects	Thrust plume (amber triangle opposite delta-v), Missile launch (ring + sparks), Missile trail (dashed orange line)
Persistent indicators	Sensor lock (dashed cyan line + ring on target), Evasive aura (pulsing blue ring), Dogfight (✂️ + red ring), Missile exhausted (×), Ion aura (pulsing blue ring, while ionRoundsLeft > 0)

3.2 Basic Mode View

In **Basic** mode there is no hex map. The screen shows **ship bento cards** grouped by faction.

Each card has three zones:

Zone	Content
Header	Ship name · faction colour dot · status badges: WRECK , DOGFIGHT , BOARDING , EVA N (evasive thrust), LOCKED (sensor locked), ION NR (ion disruption active — blue)
Hull	Hull bar (green → yellow → red) · "Hull N/M" · "Ini N"
Status (conditional)	Sensor lock → target name (with DM if set); Locked by [attacker]; inbound missiles per launcher (⚡ N× type); inbound torpedoes; launched missiles per target (🚀 N× type); reloading turrets; critical hits list; missile ammo (🚀 N/max, yellow < 25%, red at 0); sand canisters (🪨 N/max, yellow < 25%, red at 0); ion disruption (⚡ -N thrust / NR remaining)

The Status zone is hidden when none of these conditions are active.

Element	Description
DISTANCES panel	Appears above the ship list. Lists every cross-faction pair with its current range band. Use ↓ (closer) / ↑ (further) to adjust a band directly — GM override, no thrust spent.

Right-click anywhere in the background (not on a card) to open the global context menu (Roll Initiative, Add ship here).

Ships are placed at **Very Long** range by default when added to a basic mode session.

4. Phase Flow

Each combat round follows this sequence. The HUD (top-left) shows the current round and phase. Click **NEXT PHASE →** to advance.

Phase	What happens
SETUP	Place ships on the map via right-click → <i>Add ship here.</i>
INITIATIVE	Roll initiative for all ships. Sets the acting order for the round.
ACCELERATION	Each ship applies thrust to its velocity vector.
MOVEMENT	All ships move simultaneously according to their velocity vectors. (<i>Vectorial mode only — skipped in Basic mode.</i>)
ATTACK	Each ship in initiative order may attack or launch missiles.
ACTIONS	Each ship in initiative order; each crew member may perform one action.
END OF ROUND	Round counter increments. Click NEXT PHASE to begin the next round.

4.1 Phase Advance Guards

NEXT PHASE → enforces preconditions before allowing advancement:

Phase	Guard	Blocked message
All phases	No unresolved missile impacts (<code>pendingMissileImpacts</code> empty)	<i>Resolve all pending missile impacts first. Use the ↩ button in the Battle Log to re-open a dismissed impact.</i>
Setup	At least 1 ship must be placed	<i>Place at least one ship first.</i>
Initiative	Initiative must have been rolled (<code>initiativeOrder</code> non-empty)	<i>Roll initiative before advancing.</i>
Acceleration / Attack / Actions	All ships in initiative order must have acted (<code>currentActorIndex</code> ≥ <code>initiativeOrder.length</code>)	<i>N actor(s) still to act.</i>
Movement / End	Always allowed (unless missile impacts pending — see above)	—

When blocked, the button turns dim (`cursor-not-allowed`) and clicking it shows an amber warning below the button. The warning clears automatically once the condition is satisfied.

5. Setup Phase

Place ships on the hex grid before the battle begins.

5.1 Adding a Ship

Right-click any empty hex → **Add ship here**. A modal opens where you select:

- **Profile** — which saved ship profile to use.

- **Faction** — Players, Allies, Enemies, Neutral. Affects token colour and auto-roll behaviour.
- **Color** — token display colour.
- **Shape** — token silhouette: Delta, Needle, Freighter, Gunship, Cruiser, or Capital. Each shape has a distinct hull outline and bridge/cockpit detail overlay. The choice is per-placement and does not affect game mechanics.

After placing, right-click the ship token → **Assign Crew...** to review or adjust which crew member covers each role and turret (see [§ 11.3](#)).

5.2 Removing a Ship

Right-click a ship token → **Remove from battle**. Available in all phases.

***Note:** Ships start with zero velocity. Their first Acceleration phase is used to build up speed.*

6. Initiative Phase

Initiative determines the acting order for the Acceleration, Attack, and Actions phases.

Formula: `2D6 + Pilot skill + current Thrust rating + Tactics effect`
(*MgT2e CRB p. 160*)

6.1 Rolling Initiative

Right-click any hex → **Roll Initiative**. The modal shows all ships.

Ship type	Behaviour
Player ships (<code>faction =</code> <code>players</code>)	Dice inputs start empty . Enter the result of your physical 2D6 roll. The total updates live. CONFIRM is disabled until all player ships have entered dice.
NPC ships	Auto-rolled on confirm. Shown as  <i>auto</i> in the modal.
 button	Opt-in auto-roll — fills the dice fields for that ship if the player prefers the app to roll.

6.1.1 Tactics(Naval) Check (optional)

If the captain has **Tactics** ≥ 1 , an optional secondary dice row appears under the initiative row. The captain may roll $2D6 + \text{Tactics}$ — the Effect (total $- 8$, can be negative) is added to the initiative total. This check is optional; leaving it blank applies no bonus.

NPC ships with Tactics > 0 auto-roll their Tactics check on confirm.

6.2 Initiative Bonus

If the Captain uses **Improve Initiative** in the Actions phase of the previous round, the Effect is stored on the ship and added automatically to its next initiative roll — no manual input required.

The Phase Tracker (right side of screen) shows the initiative order with the current actor highlighted.

7. Acceleration Phase

Each ship adjusts its velocity vector and optionally declares evasive action.

Initiative order depends on combat mode:

- **Vectorial:** ships act in *reverse* initiative order — lowest first — so higher-initiative ships can react to slower ships' declared vectors (*TC p.174*).
- **Basic:** ships act in normal initiative order — highest first — same as the Attack and Actions steps (*CRB p.164*).

7.1 Applying Thrust

Right-click ship → **Apply Thrust**. The map enters **targeting mode**:

1. **Move the cursor** toward the hex you want the thrust to point at. A dashed line stretches from the ship to the clamped thrust endpoint.
2. **Read the overlay:**
3. Dashed line colour: **cyan** when within budget, **orange** when at cap.
4. Circle marks the thrust endpoint (ship position + delta).
5. Ghost token shows next-round position (`ship.pos + ship.vector + delta`).
6. Faint line from the inertial ghost (no thrust) to the new ghost.

7. Badge `cost / max` below the ghost.
8. **Click** to confirm. The thrust delta is applied and targeting mode exits.
9. **ESC** (or right-click) to cancel without applying thrust.

The available thrust (`thrustAvailable`) equals base thrust + any engineer bonus – thrust already used this round – M-Drive critical penalty.

7.2 Note

Full thrust is available for movement. Evasive Action is a **Reaction** declared during the Attack phase — not pre-allocated here (*CRB p.171*). See §9.9 Reactions below.

7.3 Manoeuvre (Basic Mode)

In Basic mode, **Apply Thrust** is replaced by **Manoeuvre...** in the context menu. Right-click a ship → **Manoeuvre...** during the Acceleration phase.

The modal changes the range band between the moving ship and one enemy:

Each ship manoeuvres **independently on its own initiative turn** — there is no combined thrust between ships.

Control	Description
Target	Select which enemy ship this manoeuvre targets.
↓ Approach / ↑ Flee	Direction of movement.
Thrust slider	Thrust this ship commits (0 → available thrust).
APPLY MANOEUVRE	Confirm and spend thrust — range band changes by one step if this ship's thrust meets the threshold. Disabled if thrust is below the cost.
GM SET	Override: sets the result band directly without spending thrust. Use this for initial setup and whenever thrust costs are impractical (e.g. small ships at Very Long).

*If both ships want to approach on the same round, each opens the modal on its own turn — the band closes by **two steps** in a single round.*

The threshold to change one band step depends on the current band (*MgT2e CRB p.161*):

Current band	Thrust required
Adjacent	1
Short	2
Medium	5
Long	10
Very Long	25
Distant	50

8. Movement Phase

Vectorial mode only. **Fully automatic — no player input required.**

Click **NEXT PHASE** → to execute movement. The app:

1. **Animates** every token sliding from its current position to its new position (~2 s, `easeInOut`). Input is blocked during the animation to prevent mis-clicks.
2. Advances every ship's position by its current velocity vector.
3. Detects hostile ships whose trajectories crossed within **Short range (≤ 2 hexes)** — opens the **Passing Encounter** window for each.
4. Detects ships that end in the same hex — opens the **Dogfight** engagement intent modal.

The GM watches the tokens move on the map. If no encounters or dogfights are triggered, the phase advances to **Attack** automatically.

Wrecks drift. A destroyed ship (🚢 WRECK) has no pilot and no reactor — it cannot spend thrust — but it retains its last velocity vector. Each Movement phase it continues to drift on that vector indefinitely, following Newtonian inertia. Remove it manually via right-click → Remove from battle when no longer relevant to the scenario.

8.1 Missile Impact

If one or more missile salvos reach their target's hex during the movement step, each salvo is consumed and a ⚡ **MISSILE IMPACT** modal opens automatically after the animation. Resolution follows CRB p.173 (IMPACT) in two steps.

Step 1 — Attack Roll

The attack roll happens at impact, not at launch (CRB p.173).

DM	Source
+1 per missile	Salvo size (e.g. 3 missiles → DM+3)
+2	Smart trait — all missile/torpedo weapons carry this; active only when launcher TL ≥ 9 (CRB p.79). If the launcher is sub-TL9, the +2 is not applied and the modal shows the reason.
–Pilot skill	Evasive Action (<i>optional, see below</i>)

Roll 2D6 + total DM against **8+**. Effect = total – 8.

- **Effect < 0** — MISS: modal closes, no damage applied.
- **Effect ≥ 0** — HIT: proceed to Step 2.

Evasive Action (CRB p.171): if the target has unspent thrust, the GM may click ♥ **EVASIVE ACTION** before entering the dice. This spends 1 thrust from the target's pool and applies DM –Pilot to this attack roll only. If no thrust is available, the button is disabled.

Step 2 — Damage

Roll **4D6** (Missile) or **6D6** (Torpedo) for a **single** missile/torpedo — not for the whole salvo.

Formula (*CRB p.173*):

```
net damage = max(0, roll - armour) × min(Effect, count)
```

The modal shows a full breakdown: roll / –armour / per-missile net / ×multiplier / total.

- **APPLY DAMAGE** — deducts net damage from target hull, triggers threshold criticals if applicable, logs the hit.
- **MISS / INTERCEPTED — DISMISS** — closes without applying damage (e.g. all missiles were destroyed by Point Defence before movement).

Multiple salvos impacting in the same round are resolved sequentially; the pending count is shown in the modal header. An impact sound plays when the modal opens.

Recovery

If the impact modal is accidentally dismissed before resolving, it can be re-opened at any time from the **Battle Log**: find the **< MISSILE IMPACT** entry and click the amber **↔** button at the right of the row. The impact is re-queued and the modal re-opens.

While any impact is pending, the HUD shows a pulsing **< N impacts unresolved** badge above the **NEXT PHASE →** button, and phase advance is blocked until all impacts are resolved (see §4.1).

8.2 Ships That Pass in the Night

If two hostile ships cross within **Short range (≤ 2 hexes)** during the same movement step — even if their final positions are far apart — the system detects the closest approach and opens the **Passing Encounter** window.

For each encounter the GM sees:

- Both ship names and faction colours
- Closest approach distance and range band
- **[Ship A] FIRES** — opens the Attack Modal pre-set to Ship A; button shows 
FIRED after resolving
- **[Ship B] FIRES** — opens the Attack Modal pre-set to Ship B; button shows 
FIRED after resolving

- **PASS — LET THEM GO** — dismisses the encounter immediately with no attack for either ship

Initiative order applies (TC p.177): attacks resolve at the initiative of the ship that fires first. The GM decides which ship acts first by clicking the corresponding **FIRES** button — the second ship's button remains locked until the first attack is resolved. The second ship fires only if it survives the first ship's attack; if destroyed, the Passing Encounter window closes automatically.

Both ships may fire independently. The encounter window closes automatically once both have either fired or the GM passes. Multiple encounters are resolved sequentially.

***Note:** Ships that end in the same hex trigger the **Dogfight** system instead and are handled at the movement → attack phase transition.*

In Basic mode this phase is skipped automatically — range band changes are declared during the **Acceleration phase** via the Manoeuvre modal (§7.3).

9. Attack Phase

Each ship in initiative order may make one attack or launch a missile salvo.

9.1 Attack Modal — Step 1: Config

Select the weapon to fire and the target ship. The modal shows a full DM breakdown:

- Gunner skill DM
- Weapon trait DM
- Range band DM
- Target size DM
- Evasion DM (*shown only when active*)
- Sensor Lock DM (*shown only when active*)

9.2 Attack Modal — Step 2: Roll

Target number is always **8+** (*MgT2e CRB p. 164*).

Attacker	Behaviour
Player ships	Dice inputs start empty. Enter your physical 2D6 roll. CONFIRM ROLL disabled until both dice are entered.
NPC ships	Auto-roll button. Click to roll.

Effect = Total – 8. Positive effect = hit. Effect ≥ 6 = critical hit.

9.3 Attack Modal — Step 3: Damage

On a hit, roll damage dice for the weapon. Enter or confirm the damage roll. Damage is applied to the target's hull.

9.4 Attack Modal — Step 4: Critical Hit

Triggered automatically when **Effect ≥ 6 AND damage penetrates armour** (*MgT2e CRB p.168: "it causes damage rather than just bouncing off armour"*), or when damage crosses a 10% hull threshold (Sustained Damage rule, *MgT2e CRB p. 169*).

If the weapon's damage is fully absorbed by armour ($\text{damage} - \text{armour} \leq 0$), no critical step opens even if the attack Effect is 6 or higher.

Visual effects (beam ray, impact burst, critical flash) all appear together when the critical step closes, so they are visible on the canvas.

Field	Description
Location	2D6 roll on the location table (Hull, M-Drive, J-Drive, Power Plant, Weapons, Sensors, Bridge, Fuel, Cargo, Crew, Computer).
Severity	Effect – 5, clamped 1–6. Stacks with existing criticals on the same system.
M-Drive	Sev 1 = no penalty. Sev 2–4 = –1 thrust/round. Sev 5–6 = thrust reduced to 0.

9.5 Per-Turret Firing Limit

Each turret may fire **once per round** (CRB p.164).

The Attack modal weapon list shows only turrets that have not yet fired this round, identified by their slot number (T1 , T2 ...). Once all offensive turrets have fired, the **Attack...** option disappears from the context menu until the next Attack phase or the start of a new round.

9.6 Launching Missiles

Three launcher types are available. All are selected in the Attack modal weapon list. No dice roll is required at launch — the turret is immediately marked as fired.

Missile Rack (CRB p.162)

Select **Missile Rack** , choose a target, adjust the salvo count using the **- / +** stepper (maximum capped at remaining magazine — **12 missiles per Rack**), then click  **LAUNCH SALVO** →. The button shows  **NO AMMO** and is disabled when the magazine is empty.

Missiles have **Thrust 10** and **10 rounds of guided flight**. Each round in the Movement phase they apply up to Thrust 10 toward the target's predicted position. After fuel is exhausted the salvo drifts on its last vector.

Damage: **4D6 per missile** (HG p.28).

Missile Barbette (HG p.29)

Launches a **fixed 5-missile salvo** — the count is not adjustable. Total magazine: **25 missiles** (5 salvos). Uses the same guided-flight system as the Missile Rack.

Damage: **4D6 per missile** (same as Rack — barbette ×3 multiplier does **not** apply to missile weapons; the Smart trait drives the fixed salvo size instead).

Torpedo (HG p.30–31)

Torpedoes are larger, slower-burning guided munitions. Each barbette holds **3 torpedoes**. Select **Torpedo** and fire — salvos may contain 1–3 torpedoes.

Torpedoes use the same guided-flight system as missiles. A torpedo token is rendered in red/amber on the map, separate from missile salvos.

Damage: **6D6 per torpedo** (vs 4D6 per missile) (HG p.31).

The launching turret is marked as fired after any launch. Launcher entries disappear from the Attack weapon list once ammo is empty, consistent with the per-turret limit.

9.7 Sensor Lock

Acquired via the Sensors crew action. Grants a +DM to attacks against the locked target. Shown as an **animated cyan ring** on the locked ship.

9.8 Weapon Range Limits

Each weapon has a **maximum range band** beyond which it cannot fire (*MgT2e CRB p. 167: "cannot attack targets beyond listed Range Band"*).

Weapon	Type	Max Range
Railgun	Turret	Short
Railgun Barbette	Barbette	Medium
Beam Laser	Turret	Medium
Beam Laser Barbette	Barbette	Medium
Fusion Gun	Turret	Medium
Fusion Barbette	Barbette	Medium
Plasma Gun	Turret	Medium
Plasma Barbette	Barbette	Medium
Ion Cannon	Turret	Medium
Pulse Laser	Turret	Long
Pulse Laser Barbette	Barbette	Long
Particle Beam	Turret	Very Long

Weapon	Type	Max Range
Particle Barbette	Barbette	Very Long
Missile Rack	Launcher	Special (no cap)
Missile Barbette	Launcher	Special (no cap)
Torpedo	Launcher	Special (no cap)
Sandcaster	Defensive	Special (no cap)

When a target is beyond a weapon's max range:

- An **OUT OF RANGE** label appears on the weapon button in the Config step.
- If that weapon is selected, the error is shown above the roll button.
- **ROLL ATTACK is disabled** — the shot cannot proceed until a valid weapon or a closer target is chosen.

"Special" weapons (missiles, sandcasters) have no hard range cap and are never blocked by this rule.

9.9 Reactions

Reactions are declared by the **defender** during the Attack phase, just before the attack roll (*MgT2e CRB p.171*). The app shows a ♥ **Reactions** panel in the Attack Config step as soon as a weapon and target are selected.

Each point of unspent Thrust (after movement) can be used once as a reaction. Reactions accumulate across attacks in the same round — a pulsing amber aura on the token shows when a ship has used reactions this round.

Reaction	Availability	Mechanic
Evasive Action	All attacks	Toggle button: spend 1 thrust to dodge this attack. The attack suffers DM –Pilot skill (fixed — not multiplied). (<i>CRB p.171</i>)

Reaction	Availability	Mechanic
Point Defence	Missile attacks only; target must have an unfired laser turret	Gunner (turret) check 2D6 + Gunner + laser bonus (DM+1 for 2-laser turret, DM+2 for 3-laser). Effect removes that many missiles from the salvo. Turret marked fired immediately.
Disperse Sand	Laser (Pulse/Beam) attacks only; target must have an unfired sandcaster turret	Gunner (turret) check 2D6 + Gunner. On success: +1D+Effect added to armour for this attack only. Turret marked fired immediately.

*If Point Defence destroys all missiles (count reaches 0), the attack does not proceed — click **MARK FIRED & CLOSE** to mark the attacker's turret and close.*

9.10 Special Weapon Mechanics

AP (Armour Piercing) Trait

Some weapons carry an **AP N** trait that reduces effective armour before damage.

Formula: `effectiveArmour = max(0, profile.armour - apReduction)`

Weapon	AP Value
Railgun	4
Fusion Barbette	3
Plasma Barbette	2
Railgun Barbette	5

Barbette Damage Multiplier (HG p.29)

All barbette weapons apply a **×3 damage multiplier** after armour is subtracted.

Formula: `netDamage = max(0, roll + Effect - effectiveArmour) × 3`

The multiplier applies to the net hull damage — a roll fully absorbed by armour deals zero damage regardless of the ×3. Missile and torpedo barbettes are excluded: their damage is per-projectile and uses the Smart trait salvo mechanic.

Ion Cannon (HG p.30)

The Ion Cannon does **not** deal hull damage on a hit. Instead, a successful attack applies a **thrust penalty** to the target for the remainder of the current round (and possibly longer):

Outcome	Effect
Hit (any Effect)	Roll 2D6 → <code>ionPenalty</code> applied to target's available thrust for 1 round
Hit, Effect ≥ 6	Duration extends to D3 rounds instead of 1

The penalty decays by 1 round at end-of-round via `buildNextRoundState`. When `ionRoundsLeft` reaches 0, `ionPenalty` is cleared automatically.

While active, the target's token shows a **pulsing blue aura** and the bento card shows an **ION NR** badge. The bento card's status zone shows the current penalty (`< -N thrust / NR remaining`).

```
thrustAvailable = max(0, thrust - ionPenalty - M-Drive penalty - reactionThrust)
```

10. Actions Phase — Crew

Each **crew member** may perform one action per round (*MgT2e CRB p.171*). A ship with multiple crew members can use each of them once — the modal stays open after each roll so the GM can chain actions until all crew have acted.

Right-click ship → **Crew Action**. Steps:

1. **Crew member** — only members who have not yet acted this round are listed. Already-used members are hidden until the next round.

2. **Action** — the list shows only actions that match the member's skills.
3. **Target** (*Sensor Lock only*) — select the ship to lock. **Target Salvo** (*EW — Counter Missile only*) — select the in-flight salvo to jam.
4. **Roll** — enter 2D6 for player ships; NPC ships have a 🎲 auto-roll button.

After the roll the modal shows the result (SUCCESS / FAILED + Effect). Click **ANOTHER ACTION** to act with a second crew member, or **CLOSE** to exit.

10.1 Skill DM Override

When an action is selected, the relevant skill level is pre-filled as the roll DM. The GM can override this value for specialisations (e.g. Engineer(M-Drive) 3 vs. generic Engineer 2). The ↺ button resets to the base skill level.

10.2 Available Actions

All checks are **2D6 + skill DM vs. 8+** unless marked Automatic.

Role	Action	Difficulty	Effect on success
Captain	Improve Initiative	8+ (Leadership)	+Effect added to this ship's initiative roll next round (<i>CRB p.166</i>)
Engineer	Overload M-Drive	8+ (Engineer)	+Effect Thrust available this round (<i>CRB p.167</i>)
Engineer	Repair System	8+ (Engineer)	Removes 1 critical hit from this ship (<i>CRB p.167</i>)
Gunner	Reload Turret	Automatic	Reloads 1 missile turret; no roll required (<i>CRB p.167</i>)
Sensors	Sensor Lock	8+ (Electronics)	DM+2 flat to all attacks against the selected target (<i>CRB p.172</i>)
Sensors	Electronic Warfare	8+ (Electronics)	Removes an enemy sensor lock from this ship (<i>CRB p.167</i>)
Sensors	EW — Counter	10+ (Electronics)	Removes Effect missiles (min 1) from one in-flight salvo. Cumulative

Role	Action	Difficulty	Effect on success
	Missile		across rounds; a salvo may only be EW'd once per round (<i>CRB p.173</i>)

A skill level of 0 in a role grants **no actions** for that role. Skills with level ≥ 1 unlock the role's full action list.

NPC ships resolve all non-automatic rolls automatically when the GM clicks  EXECUTE ACTION.

11. Crew System

Ships have a list of **named crew members**, each with individual skill ratings.

11.1 Skills

Abbrev	Skill	Used for
PLT	Pilot	Initiative roll, evasion DM, dogfight/pursuit checks
LDR	Leadership	Improve Initiative action (Actions phase)
TAC	Tactics	Initiative DM at start of battle (Initiative phase)
ENG	Engineer	Overload M-Drive action, Repair System action
GNR	Gunner	Attack DM, Reload Turret action
SEN	Sensors	Sensor Lock action, Electronic Warfare action, EW — Counter Missile action

Skill levels range from **0 to 5**.

11.2 Multi-Skill Members

One crew member can hold **multiple skills**. A solo pilot/gunner on a fighter has both Pilot 2 and Gunner 2 on the same crew entry.

11.3 Crew Role Assignments

Before battle, each role slot must be filled by an assigned crew member. Roles are **optional** — an unassigned role contributes **0 skill** (no bonus).

Right-click any ship token → **Assign Crew...** to open the assignment modal.

Using the Assign Crew modal

The modal is divided into two sections:

Roles — one dropdown per non-gunner role:

Slot	Dropdown label
Pilot	Pilot
Leadership	Leadership (LDR)
Tactics	Tactics (TAC)
Engineer	Engineer
Sensors	Sensors

Gunners — one dropdown per turret slot (T1, T2...), with the weapon names listed next to the slot label.

Each dropdown lists all named crew members. The relevant skill level is shown in brackets — e.g. `Mira Vasquez [pilot 1]` or `Joko Hendrik [no skill]`. Select `– unassigned –` to leave the slot empty.

Buttons:

- **CLEAR ALL** — resets every slot to unassigned.
- **SAVE ASSIGNMENTS** — commits the selection to the ship and closes the modal.

If the ship profile has no named crew, the modal shows "No named crew on this ship. Add crew members in the profile editor."

Effect of unassigned roles

Role slot	Effect if unassigned
Pilot	Initiative roll, evasion DM, dogfight use skill 0
Leadership	Improve Initiative action not available
Tactics	No Tactics(naval) check at initiative
Engineer	Engineer actions use skill 0
Gunner (T1, T2...)	That turret cannot fire this session
Sensors	Sensors actions use skill 0

A crew member can be assigned to multiple slots (e.g., the same person as both Pilot and Gunner T1 on a light fighter).

When a ship is placed on the map the app auto-assigns the best-skilled member per role as a starting point. The GM can adjust at any time before or during combat.

***NPC ships** without explicit crew assignments fall back to the highest skill across all crew members (backward-compatible behaviour preserved).*

11.4 Editing Crew

In the ship profile form, use **+ ADD CREW** to add a member and **✕** to remove one. Each row has a name field and compact skill inputs.

Default profiles (Free Trader, Scout, Patrol Cruiser...) and catalog ships (High Guard 2022 catalog) come with pre-generated crew — names and skill levels chosen as a reasonable starting point. **These are fully editable**: open the profile in the Dashboard →  **Edit** → **Crew Manifest** to rename members, adjust skill levels, add specialists, or remove crew that doesn't fit the scenario. Changes take effect the next time the ship is placed in battle.

***Example:** Replace Cass Oduya (Gunner 1) on the Free Trader with your player character's name and actual Gunner skill before the session starts.*

Legacy profiles using the old flat object format (`{pilot: 2, gunner: 1}`) are automatically migrated to the named array format when loaded in the form.

12. Undo / Redo

Every user action that changes game state pushes a snapshot to the undo stack (capped at **20 entries**).

Control	Action
 Undo	Restore the previous state. Appears in HUD when stack is non-empty. Shortcut: <code>Ctrl+Z</code> / <code>Cmd+Z</code> .
 Redo	Re-apply an undone action. Appears in HUD when redo stack is non-empty. Shortcut: <code>Ctrl+Y</code> / <code>Cmd+Shift+Z</code> .

Both buttons are hidden when their respective stacks are empty — they appear only when relevant.

*The battle log is **not** rolled back on undo. Instead, a  **Undo** entry is appended to the log so the action history remains readable.*

13. Save & Resume

13.1 Autosave

The app autosaves to IndexedDB after every significant action (ships added/removed, damage applied, phase advanced, etc.). No manual trigger needed.

On next visit, the  **RESUME AUTOSAVE** button appears on the Dashboard with round, phase, and ship count.

13.2 Manual Save

Click  **SAVE** in the HUD at any time to download the full session as a `.json` file.

13.3 Resume from File

On the Dashboard, click ↓ **RESUME FROM FILE** and select your saved `.json` file. A preview screen shows the full roster (name, faction, hull, position) before you confirm

loading.

13.4 Profile Export / Import

Ship profiles are separate from battle sessions. Use ↑ **EXPORT** and ↓ **IMPORT** in the profile panel to share or back up profiles independently.

Clicking 🏠 in the HUD returns to the Dashboard. A confirmation modal warns that unsaved battle data will be lost — save first if you need to resume.

14. Dogfight

MgT2e CRB p.138 — Vehicle Combat / Dogfighting rules adapted for space combat.

14.1 What is a Dogfight?

A **dogfight** is a close-range sub-system that activates automatically when two or more ships from different factions end the **Movement phase in the same hex** (vectorial mode only).

Standard combat is suspended for the involved ships. Instead, the round subdivides into **6 micro-rounds** of 6 seconds each — one full standard round equals six micro-rounds of dogfight.

14.2 Engagement

When the app detects a potential dogfight at the end of Movement, the **CONFIRM INTENTS** modal opens.

For each ship the GM declares:

Intent	Outcome
Both YES	Dogfight activates immediately
Both NO	Ships treated as Short Range (distance 1) — no dogfight
Mixed	Pursuit check required (see §14.3)

14.3 Pursuit Check

Formula: **2D6 + Pilot + Tonnage DM + free Thrust**

- **Free Thrust** = profile thrust – thrust used this round
- **Tonnage DM**: <50t → 0; 50–99t → –1; 100–199t → –2; +–1 per 100t above 100

If the pursuer's total exceeds the evader's total → dogfight activates. Otherwise → Short Range, no dogfight.

Enter dice manually for each side. The app computes and compares totals live.

14.4 Micro-Round Flow

Open the round from the **HUD dogfight tracker** ( DOGFIGHT panel, top-left).

Step 1 — Declare escape (optional)

At the start of each micro-round the GM may declare that a ship wants to flee.

- If its thrust exceeds all enemy thrusts → **auto-escape**, no check needed.
- If enemies choose not to pursue → **auto-escape** (toggle "NOT PURSUING").
- Otherwise → pursuit check (same formula as §14.3).

Step 2 — Pilot check

Each remaining ship rolls **2D6 + Pilot + Tonnage DM + Thrust + previous round bonus**.

DM	Source
Pilot skill	<code>getCrewSkill(crew, 'pilot')</code>
Tonnage DM	See §14.3 table
Thrust	Profile thrust – thrust used this round
Extra enemies	–(number of enemy ships – 1) when outnumbered
Round bonus	Previous round winner's margin carries forward as a +DM

Step 3 — Result

Outcome	Effect
Winner	+2 DM to all attacks this micro-round; chooses enemy fire arc
Loser	-2 DM to all attacks
Tie	Fixed weapons cannot fire; turrets OK; no positional DM

Apply the attack DMs shown in the modal when opening the **Attack** panel.

Step 4 — Advance

Click **ADVANCE** → **MICRO-ROUND N+1/6**. After micro-round 6 the dogfight ends automatically and all ships return to normal combat flow.

14.5 Token Visuals

Ships in a dogfight display:

- **Pulsing amber ring** around the token
-  **badge** top-right of the token
- Ghost position and velocity arrow are hidden (no movement during dogfight)

14.6 Escape Mid-Dogfight

Escape can be declared at the start of any micro-round (Step 1 above). Conditions:

```
Auto-escape: ship.thrust > max(enemy thrusts)
              OR enemies choose not to pursue
Check:       2D6 + Pilot + Tonnage DM + free Thrust (same as §14.3)
              evader total > pursuer total → escaped
```

On successful escape `inDogfight` is cleared; the ship re-enters normal combat from the next standard round.

Technical reference: *dogfight-system-design.md*

15. Boarding

HG 2022 pp.125–135 — full 4-phase boarding system.

Boarding is a sub-system that activates when an attacker moves adjacent to a target and meets the thrust requirement. Combat shifts from the hex map to the interior of the target ship.

15.1 Triggering a Boarding Action

Right-click the **attacker ship** and select  **Board [target name]...**

The option is visible only when:

```
distance(attacker, target) ≤ 1 (Adjacent or Close)
AND target.inDogfight === null (target not in an active dogfight)
AND attacker.thrust ≥ target.thrust
    OR target M-Drive critical is disabled
AND different factions
```

The **Boarding Setup** modal opens. Select the target and confirm. The boarding moves immediately to Phase 2 — Contact.

15.2 Phase 1 — Approach

Handled outside the app. The approach phase is the normal combat movement that brought the two ships together. The GM declares the boarding when conditions are met.

Voluntary boarding (target cooperates): skip to Contact immediately.

Forced boarding: if the target attempted to flee and failed (or is immobilised), proceed to Contact.

15.3 Phase 2 — Contact

The  **CONTACT** modal opens. The GM selects the entry method:

Method	Check	Difficulty	Time	DM
Airlock (cooperative)	None	—	Instant	—
Airlock (forced)	Mechanic (STR)	14+	2D rounds + 1D	—
Maintenance Hatch	Mechanic (STR)	12+	2D rounds	— 
Breaching Tube	None	—	< 2 min	—
Forced Linkage Apparatus	Pilot (DEX)	8+	Immediate	+2
Hull Cut	Mechanic (DEX)	8+/round	Per round	— 

 = decompression risk if compartment not evacuated.

Modifiers:

-  **Tumbling** — defender rotating the ship: DM -1 to all Contact checks
-  **Forced Linkage** — DM +2 to all Contact checks; defender cannot manoeuvre

Hull Cut tracker: select component (Hatch / Airlock / Hull) and cutting tool, roll each round. Damage reduces component Resilience; breach achieved when damage ≥ breach threshold.

When entry is secured, click **ADVANCE TO CONFLICT** → .

15.4 Phase 3 — Conflict

The  **CONFLICT** modal tracks the boarding fight.

Tactical objectives — check each when captured:

Objective	Effect when captured
Bridge	Remote control of all ship systems disabled for enemy

Objective	Effect when captured
Engineering	Propulsion, reactor, life support under attacker control
Turrets	Weapon systems under attacker control

The ship is considered taken when all three objectives are captured.

Combat tools:

- **ROLL STACKING** — roll 2D ≥ 10 to target a combatant beyond the first in a corridor (HG p.131)
- **ROLL MISSED SHOT** — roll 2D on the missed-shot table for every attack that misses; optional **Armored bulkhead (DM -1)** toggle

Weapon DM in tight spaces: Rifles -2 · Heavy weapons -4 · Grenades → automatic 6D+

When the fight is resolved, click **END CONFLICT — ADVANCE TO SECURITY** → .

15.5 Phase 4 — Security

The  **SECURITY — BOARDING OUTCOME** modal resolves the action.

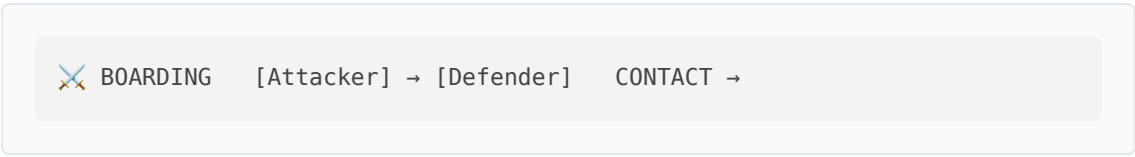
Outcome	Effect
Attacker wins	Boarding party controls the ship; optional faction transfer
Defender repels	Boarders eliminated, captured, or driven off
Ship destroyed	Target destroyed by internal damage during conflict

If **Attacker wins** and faction transfer is enabled, the captured ship's faction changes to match the attacker. Enemy crew is removed from the roster.

Click **CONFIRM OUTCOME** to close the boarding.

15.6 HUD Indicator

While a boarding is active, the HUD shows a  **BOARDING** badge below the standard tracker:



Click the phase button to reopen the relevant modal at any time.

15.7 Boarding and Normal Combat

- Ships **inBoarding** do not participate in the standard Attack phase
- A ship in an active **dogfight** cannot be targeted for boarding — it must exit the dogfight first
- A ship with **Forced Linkage active** cannot use thrust to manoeuvre
- If the target is destroyed during Conflict, resolve with outcome **Ship destroyed**
- Normal rounds continue in parallel — the GM can advance phases and resolve the boarding on its own timeline, as with dogfights

Technical reference: *boarding-system-design.md*

16. Further Reading

Document	Contents
thrust-and-drift-space-combat-simulator-spec.md	Full technical spec — data models, store actions, component structure, roadmap
dogfight-system-design.md	Dogfight sub-system design — micro-round flow, pursuit checks, escape mechanics
boarding-system-design.md	Boarding sub-system design — entry methods, hull-cut, conflict objectives
obstacles-system-design.md	Environmental obstacles design — asteroid field, gravity well, debris, nebula

Document	Contents
weapons-expansion-design.md	Weapons expansion design — barbette ×3 multiplier, AP trait, Ion Cannon penalty, Torpedo/Missile Barbette ammo, sandcaster canister tracking

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